



Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 × 10 = 10]

- Write the reagents of nitration of benzene.
- Write the value of quantum numbers n , l and m for $2s$ orbital.
- MRI is the application of which spectroscopic method?
- London forces or dispersion forces operates between _____.
- How the entropy of the system changes when water is frozen?
- What is Hard-soft acid base principle?
- If two stereoisomers of tartaric acid have $2S$, $3R$ and $2S$, $3S$ configurations, then they are _____.
- What are the conditions of Cannizzaro reaction?
- State whether naphthalene is aromatic, antiaromatic or nonaromatic and give reason.
- Write the equation of rotational constant B of a molecule.
- What is the value of critical coefficient?
- Write the half-cell representation of saturated calomel electrode.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 × 3 = 15]

- What is the role of Lewis acid in halogenation of benzene? Write the structure of the substrate which on ozonolysis will provide acetone as the only product. [5]
- Write the significance of van-der Waal's constants. [5]
- What are conformational isomers? Explain by taking ethane as an example. Draw the potential energy diagram of ethane. [5]
- Prove that for a constant pressure process, where work is only mechanical, the heat absorbed by the system (Q_p) is equal to the increase in enthalpy (ΔH). <https://www.makaut.com> [5]
- NO is paramagnetic while NO^+ is diamagnetic. Justify using MO diagram. Write electronic configuration of NO . [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 × 3 = 45]

- Define 'Standard cell' with example. [3]
 - Find the relation between ΔH and ΔU for an ideal gas undergoing a P-V type mechanical work as follows:
 $\Delta H = \Delta U + \Delta nRT$ [4]
 - Discuss any one statement of First law of thermodynamics and explain it mathematically. [4]
 - Explain 'Galvanic cell corrosion'. [4]
- Demonstrate the shape of the following compounds.
(i) ClF_3 (ii) XeF_2 (iii) BrF_5 (iv) SCl_6 [8]
 - Arrange NH_3 , H_2O , CH_4 according to increasing bond angle stating the reason. [4]
 - Dipole moment of BF_3 is zero while that of NF_3 is $0.24D$. Justify. [3]
- Derive Schrodinger Equation. [5]
 - Write the conditions of acceptable wave function ψ . [5]
 - Write short note on extrinsic semiconductor and its types. [5]
- What will be the product/ products if $CH_3CH=CH_2$ is reacted with HBr ? Show the steps of reaction. [5]

- (b) What change will be observed in the above reaction if peroxide is added along with HBr? { 2 }
- (c) Choose the more suitable substrate of E1 reaction between tertiary-butyl chloride and methyl chloride. Give reason. { 3 }
- (d) Explain the stereo chemical aspects of SN^1 and SN^2 reactions with suitable examples. { 5 }
11. (a) Calculate the CFSE and Magnetic moment of $K_3[FeF_6]$. { 5 }
- (b) Calculate the CFSE of both high spin and low spin complexes of d^7 and d^4 complexes. { 5 }
- (c) Give the molecular energy level diagram of CO. Comment on magnetic behaviour and bond order. { 5 }

*** END OF PAPER ***